

Atomic Structure

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Protons are positive and in the nucleus, neutrons have no charge and are in the nucleus, and electrons are negative and outside the nucleus.
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Round 1 - Retrieve It

1. What charge does a proton have, and where is it found?

2. What charge does a neutron have, and where is it found?

3. What charge does an electron have, and where is it found?

4. In a neutral atom, how do the numbers of protons and electrons compare?

Round 2 - Sketch It

Draw a simple model of a neutral carbon atom with 6 protons, 6 neutrons, and 6 electrons. Label the nucleus and electron region.



Atomic Structure - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. An atom has 8 protons, 8 neutrons, and 10 electrons. What is its overall charge? Explain.

2. Two atoms each have 6 protons. One has 6 neutrons and the other has 8. Are they the same element?

3. If an atom loses one electron, does its nucleus change? What happens to its charge?

Round 4 - Reason from Evidence

Explain why changing the number of neutrons can make a different isotope, while changing the number of protons makes a different element.

Atomic Structure - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Atomic Structure in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect atomic structure to the periodic table. Which particle determines an element's atomic number?

E**Extend**

A mystery atom has 13 protons and 10 electrons. Predict its charge and explain what evidence you used.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Atomic Structure

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Positive; in the nucleus.
2. No charge; in the nucleus.
3. Negative; outside the nucleus in the electron cloud.
4. They are equal.

Sketch It - look for

A nucleus labeled with 6 protons and 6 neutrons, with 6 electrons shown outside the nucleus. Exact shell placement is not required unless taught.

Apply It

1. 2- because it has two more negative electrons than positive protons.
2. Yes. Element identity is set by the number of protons; these are isotopes of the same element.
3. The nucleus does not change; the atom becomes more positive by one charge.

Reason from Evidence - look for

Look for: element identity depends on proton number; neutron number can vary within the same element; changing protons changes atomic number and therefore the element.

ACE - Extend look-for

Atomic number comes from protons. Mystery atom charge is 3+ because 13 positive protons outweigh 10 negative electrons by three.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Elements, Compounds & Mixtures

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round

1. Check the card only after you commit to an answer.

Cue focus

An element contains one kind of atom, a compound has elements chemically bonded in fixed combinations, and a mixture contains substances together without chemical bonding.

Round 1 - Retrieve It

1. What makes a substance an element?

2. What makes a substance a compound?

3. What makes a sample a mixture?

4. Which can usually be separated by physical methods: a compound or a mixture?

Round 2 - Sketch It

Create three particle sketches: one element, one compound, and one mixture. Use circles of different patterns or labels to show different atom types.



Elements, Compounds & Mixtures - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Classify oxygen gas, O₂: element, compound, or mixture?

2. Classify carbon dioxide, CO₂: element, compound, or mixture?

3. Classify saltwater: element, compound, or mixture?

Round 4 - Reason from Evidence

A student says, "If two kinds of atoms are present, the sample must be a compound." Explain why that statement is incomplete.

Elements, Compounds & Mixtures - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Elements, Compounds & Mixtures in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect particle diagrams to classification. What visual evidence would show that two substances are mixed but not bonded?

E**Extend**

You are given iron filings and sand. Propose a physical method to separate the mixture and explain why it works.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Elements, Compounds & Mixtures

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Retrieve It

1. It contains only one kind of atom.
2. Two or more different elements are chemically bonded in a fixed ratio.
3. Two or more substances are physically combined but not chemically bonded.
4. A mixture.

Sketch It - look for

Element: one atom type only. Compound: identical bonded groups containing at least two atom types. Mixture: two or more particle types together without all being chemically bonded into one repeating substance.

Apply It

1. Element, because it contains only oxygen atoms.
2. Compound, because carbon and oxygen are chemically bonded.
3. Mixture, because salt and water are physically combined and can be separated physically.

Reason from Evidence - look for

A mixture can also contain more than one kind of atom or molecule. A compound requires different elements to be chemically bonded in a fixed composition.

ACE - Extend look-for

Use a magnet to remove the iron because magnetism is a physical property; the substances are not chemically bonded.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Metals, Nonmetals & Metalloids

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Metals are usually shiny and conductive, nonmetals are often dull and poor conductors, and metalloids have mixed properties near the periodic table stair-step.
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Round 1 - Retrieve It

1. Name two common properties of metals.

2. How do nonmetals usually compare with metals in electrical conductivity?

3. Where are many metalloids found on the periodic table?

4. Why are metalloids useful in electronics?

Round 2 - Sketch It

Draw a simplified periodic table rectangle. Mark a stair-step line and label the general metal, nonmetal, and metalloid regions.



Metals, Nonmetals & Metalloids - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A sample is shiny, bends without breaking, and conducts electricity well. Best classification?

2. A sample is dull, brittle, and a poor conductor. Best classification?

3. A sample has some metallic shine but only conducts under certain conditions. Best classification?

Round 4 - Reason from Evidence

A device needs a material that can control electrical flow rather than conduct extremely well or block it completely. Which category is the best starting point, and why?

Metals, Nonmetals & Metalloids - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Metals, Nonmetals & Metalloids in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect properties to use. Why is copper useful for electrical wiring while many nonmetals are used as insulators?

E**Extend**

An unknown solid is brittle but has moderate conductivity. Predict its category and name one additional test that would strengthen your claim.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Metals, Nonmetals & Metalloids

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Retrieve It

1. Examples: shiny/lustrous, conductive, malleable, ductile.
2. Nonmetals are generally poor conductors.
3. Along the stair-step boundary between metals and nonmetals.
4. They have intermediate or semiconducting electrical properties.

Sketch It - look for

Metals occupy most of the left/center, nonmetals are mainly upper right, and metalloids lie along the stair-step boundary. Exact element placement is not required for this sketch.

Apply It

1. Metal.
2. Nonmetal.
3. Metalloid.

Reason from Evidence - look for

Metalloid, because semiconducting behavior falls between strong conductors and insulators.

ACE - Extend look-for

Likely metalloid. Additional evidence could include location on the periodic table, luster, or measured conductivity compared with known samples.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

States of Matter

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Solids have fixed shape and volume, liquids have fixed volume but take the container shape, and gases have neither fixed shape nor fixed volume because particles are farther apart and move more freely.
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Round 1 - Retrieve It

1. Which state has a fixed shape and fixed volume?

2. Which state has a fixed volume but takes the shape of its container?

3. Which state has neither fixed shape nor fixed volume?

4. Which state usually has the greatest particle spacing?

Round 2 - Sketch It

Draw particle models for a solid, liquid, and gas in equal-size boxes. Show relative spacing and arrangement.

States of Matter - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Why can air in a syringe be compressed more easily than water?

2. Name the change from solid to liquid and from liquid to gas.

3. Name the change from gas to liquid and from liquid to solid.

Round 4 - Reason from Evidence

Explain what happens to particle motion and spacing as matter is heated from solid to liquid to gas.

States of Matter - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain States of Matter in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect state of matter to particle motion. Why does a liquid flow while a solid keeps its shape?

E**Extend**

Predict what happens to the particles of water vapor as it cools enough to condense on a cold surface.

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TEACHER KEY - States of Matter

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Retrieve It

1. Solid.
2. Liquid.
3. Gas.
4. Gas.

Sketch It - look for

Solid particles close and ordered; liquid particles close but disordered/mobile; gas particles much farther apart.

Apply It

1. Gas particles have much more empty space between them.
2. Melting; vaporization/evaporation.
3. Condensation; freezing.

Reason from Evidence - look for

Particles gain kinetic energy, move faster, and become less constrained; spacing generally increases, especially in the gas state.

ACE - Extend look-for

Particles lose energy, slow down, move closer together, and form liquid water.

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Physical Properties of Matter

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Mass, volume, density, solubility, conductivity, and magnetism are physical properties that can be observed or measured without changing a substance into something new.
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Round 1 - Retrieve It

1. What does mass measure?

2. What does volume measure?

3. What is the formula for density?

4. Name three other physical properties on the cue card.

Round 2 - Sketch It

Make a six-part property wheel. Label mass, volume, density, solubility, conductivity, and magnetism. Add one quick test or tool for each.

Physical Properties of Matter - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A rock has mass 54 g and volume 20 mL. Calculate its density.

2. Which property is most useful for separating iron filings from sand?

3. Which property could help compare copper wire and plastic for use in a circuit?

Round 4 - Reason from Evidence

A student says, "If I measure a property, I always change the substance." Use two physical-property examples to show why that is incorrect.

Physical Properties of Matter - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Physical Properties of Matter in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect density to floating and sinking. What additional information would you need to predict whether an object floats in a liquid?

E**Extend**

Design a two-test plan to distinguish two unlabeled solids using only physical properties.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Physical Properties of Matter

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Retrieve It

1. The amount of matter in a sample.
2. The amount of space a sample occupies.
3. Density = mass / volume.
4. Any three: solubility, conductivity, magnetism, plus mass, volume, density.

Sketch It - look for

Examples: balance for mass; graduated cylinder/ruler for volume; mass/volume calculation for density; dissolve test for solubility; circuit/tester for conductivity; magnet for magnetism.

Apply It

1. 2.7 g/mL.
2. Magnetism.
3. Electrical conductivity.

Reason from Evidence - look for

Measuring mass, volume, density, conductivity, magnetism, or solubility can identify a property without creating a new substance. Some tests may change form or location but not chemical identity.

ACE - Extend look-for

Accept reasonable plans using measurable properties such as density, magnetism, conductivity, or solubility, with clear expected evidence.

Suggested use

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Chemical Formulas

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	A subscript tells how many atoms of one element are in a formula unit, while a coefficient multiplies every atom in the entire formula.
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Round 1 - Retrieve It

1. What does a subscript tell you?

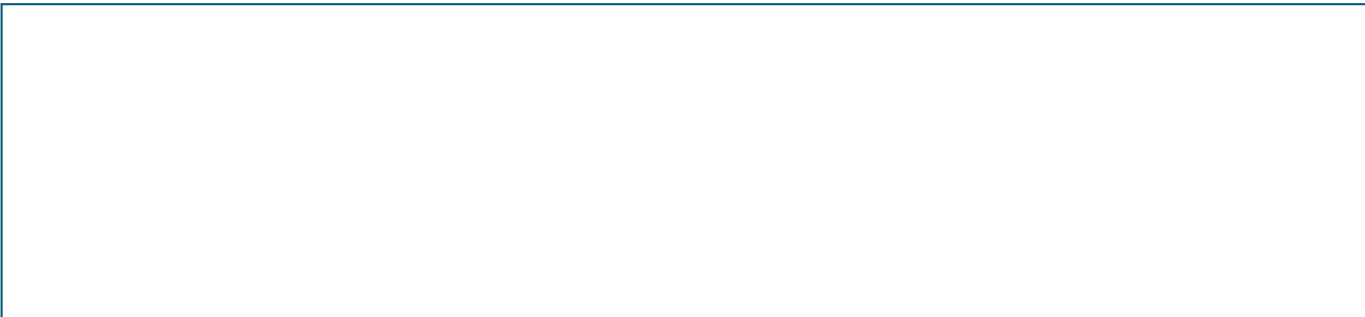
2. What does a coefficient tell you?

3. In H_2O , how many H atoms and O atoms are represented?

4. In $3\text{H}_2\text{O}$, how many water molecules are represented?

Round 2 - Sketch It

Use circles labeled H and O to draw one H_2O molecule, then draw what $3\text{H}_2\text{O}$ represents.



Chemical Formulas - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Count atoms in $3\text{H}_2\text{O}$.

2. Count atoms in 2CO_2 .

3. Count atoms in Al_2O_3 .

Round 4 - Reason from Evidence

Compare $2\text{H}_2\text{O}$ and H_4O_2 . They show the same total atom counts. Why do the symbols not mean the same thing chemically?

Chemical Formulas - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Chemical Formulas in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect coefficients to conservation of atoms in chemical equations. Why do chemists adjust coefficients rather than subscripts when balancing?

E**Extend**

Write the total number of each atom in 4NH_3 and explain your multiplication.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Chemical Formulas

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Retrieve It

1. The number of atoms of the element immediately before it.
2. How many whole molecules or formula units are present; it multiplies every atom in the formula.
3. 2 H and 1 O.
4. 3 molecules.

Sketch It - look for

One H_2O : two H atoms bonded with one O atom. $3\text{H}_2\text{O}$: three separate H_2O groups, totaling 6 H and 3 O atoms.

Apply It

1. 6 H and 3 O; 9 atoms total.
2. 2 C and 4 O; 6 atoms total.
3. 2 Al and 3 O; 5 atoms total.

Reason from Evidence - look for

A coefficient changes the number of intact H_2O molecules. Changing subscripts changes the formula/ratio within one substance and can represent a different substance.

ACE - Extend look-for

N = 4 and H = 12 because the coefficient 4 multiplies the entire NH_3 formula.

Suggested use

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Chemical Changes

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Evidence of a chemical change can include color change, gas formation, precipitate formation, and temperature or light change when those changes result from a reaction.
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Round 1 - Retrieve It

1. Name four common signs that a chemical reaction may have occurred.

2. What is a precipitate?

3. Does every color change prove a chemical reaction?

4. Why is boiling water not automatically evidence of a chemical change?

Round 2 - Sketch It

Create a four-box "reaction evidence" chart for color, gas, precipitate, and temperature/light. Add a small example or symbol to each.

Chemical Changes - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A metal rusts and forms a reddish-brown coating. Physical or chemical change?

2. Ice melts into liquid water. Physical or chemical change?

3. Two clear solutions are mixed and a solid appears. What evidence is present?

Round 4 - Reason from Evidence

Why is it stronger to use more than one piece of evidence before claiming a chemical reaction occurred?

Chemical Changes - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Chemical Changes in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect chemical changes to conservation of mass. If new substances form, what should happen to total mass in a closed system?

E**Extend**

A sealed bag becomes cold and inflates after two substances are mixed. What evidence suggests a reaction, and what additional observation would you seek?

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Chemical Changes

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Retrieve It

1. Color change, gas forms, precipitate forms, temperature or light change.
2. A new solid that forms from substances in solution.
3. No. It is evidence that must be interpreted with the situation.
4. The gas is still water; no new substance is formed.

Sketch It - look for

Any accurate, labeled examples showing the four evidence categories. Examples should distinguish reaction evidence from simple phase change where possible.

Apply It

1. Chemical change; a new substance forms and color changes.
2. Physical change; the substance remains H₂O.
3. Precipitate formation, strong evidence of a chemical reaction.

Reason from Evidence - look for

Single observations can have nonchemical explanations. Multiple consistent signs strengthen the claim that new substances formed.

ACE - Extend look-for

Temperature change and gas production are evidence. Additional evidence might include color change, precipitate, or comparing reactants/products while ensuring the inflation is not just trapped air.

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Net Force

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Forces in the same direction add, forces in opposite directions subtract, and net force is the overall force with a magnitude and direction.
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Round 1 - Retrieve It

1. How do you combine forces that act in the same direction?

2. How do you combine forces that act in opposite directions?

3. What unit is commonly used for force?

4. What does a net force of 0 N mean about the forces?

Round 2 - Sketch It

Draw a box with a 10 N arrow to the right and a 4 N arrow to the left. Add a net-force arrow and label it.

Net Force - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Two students push a cart right with 5 N and 7 N. Net force?

2. A rope is pulled 18 N east and 11 N west. Net force?

3. A box has 9 N left and 9 N right. Net force?

Round 4 - Reason from Evidence

A cart has 12 N east, 7 N west, and another 5 N west. Find the net force and predict whether the forces are balanced.

Net Force - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Net Force in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect net force to acceleration. What should happen when the net force is not zero?

E**Extend**

Create your own three-force example with a net force of 4 N left. Show the arithmetic.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Net Force

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Retrieve It

1. Add their magnitudes.
2. Subtract the smaller magnitude from the larger and use the direction of the larger force.
3. Newton, N.
4. They are balanced.

Sketch It - look for

Net force is 6 N to the right.

Apply It

1. 12 N right.
2. 7 N east.
3. 0 N; balanced forces.

Reason from Evidence - look for

12 N east versus 12 N west gives 0 N; forces are balanced.

ACE - Extend look-for

Many answers possible. Check that signed/directional forces combine to 4 N left.

Suggested use

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Balanced & Unbalanced Forces

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Balanced forces cancel so motion does not change, while unbalanced forces produce a nonzero net force and change motion.
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Round 1 - Retrieve It

1. What is the net force when forces are balanced?

2. Can an object be moving while forces are balanced?

3. What is true about net force when forces are unbalanced?

4. What can an unbalanced force change?

Round 2 - Sketch It

Draw one balanced-force example and one unbalanced-force example. Use arrows with numbers and label the net force for each.

Balanced & Unbalanced Forces - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A book rests on a table. Gravity pulls down and the table pushes up equally. Balanced or unbalanced?

2. A car moves straight at constant speed. Are the overall forces necessarily unbalanced?

3. A soccer ball speeds up after being kicked. Balanced or unbalanced during the speed change?

Round 4 - Reason from Evidence

Explain why "balanced forces" does not mean "the object is not moving."

Balanced & Unbalanced Forces - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Balanced & Unbalanced Forces in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect balanced forces to Newton's first law. What stays the same when net force is zero?

E**Extend**

A skateboarder rolls at constant speed, then begins slowing. Describe how the force situation changed.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Balanced & Unbalanced Forces

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Retrieve It

1. 0 N.
2. Yes. It can move at constant speed in a straight line.
3. Net force is not zero.
4. An object's speed, direction, or both.

Sketch It - look for

Balanced example has equal opposing forces and 0 N net. Unbalanced example has unequal forces and a nonzero net force in the stronger direction.

Apply It

1. Balanced.
2. No. Constant velocity is consistent with balanced forces.
3. Unbalanced.

Reason from Evidence - look for

Balanced forces mean zero acceleration, not zero velocity. An object can remain at rest or keep moving at constant velocity.

ACE - Extend look-for

Initially net force was approximately zero. While slowing, an unbalanced force acts opposite the motion.

Suggested use

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Newton's Laws of Motion

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus Newton's first law describes inertia, the second law connects force, mass, and acceleration ($F = m \times a$), and the third law describes equal and opposite interaction forces.

Round 1 - Retrieve It

1. What is the big idea of Newton's first law?

2. Write Newton's second-law relationship.

3. What is the big idea of Newton's third law?

4. What happens to acceleration if the same force acts on a larger mass?

Round 2 - Sketch It

Draw a simple scene for each law: inertia, $F = m \times a$, and an action-reaction interaction. Label 1, 2, and 3.



Newton's Laws of Motion - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A 3 kg cart accelerates at 4 m/s^2 . What net force acts on it?

2. A passenger lurches forward when a car stops suddenly. Which law best explains this?

3. A swimmer pushes water backward and water pushes the swimmer forward. Which law?

Round 4 - Reason from Evidence

Students sometimes say third-law forces cancel each other. Explain why the pair does not cancel on one object.

Newton's Laws of Motion - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Newton's Laws of Motion in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect Newton's second law to net force. Which force value belongs in $F = m \times a$ when several forces act?

E**Extend**

Two carts get the same 20 N net force. One is 2 kg and one is 5 kg. Predict which accelerates more and calculate both accelerations.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Newton's Laws of Motion

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Retrieve It

1. Objects resist changes in motion; motion changes only when a net external force acts.
2. $F = m \times a$.
3. For every interaction force, there is an equal-magnitude, opposite-direction force on the other object.
4. Acceleration decreases.

Sketch It - look for

Any accurate examples. Third-law pair must act on two different objects.

Apply It

1. 12 N.
2. First law/inertia.
3. Third law.

Reason from Evidence - look for

The equal and opposite forces act on different objects. Forces cancel only when forces on the same object sum to zero.

ACE - Extend look-for

2 kg cart: 10 m/s². 5 kg cart: 4 m/s². Smaller mass accelerates more with the same net force.

Suggested use

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Friction & Surface Type

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Friction acts opposite relative motion; rougher surfaces usually produce more friction and smoother surfaces usually produce less.
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Round 1 - Retrieve It

1. Which direction does friction act compared with motion?

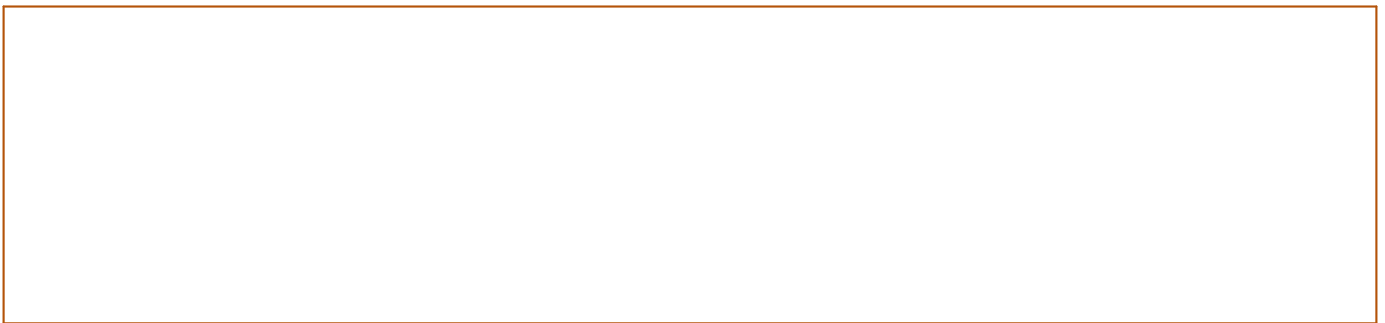
2. Which usually produces more friction: a rough surface or a smooth surface?

3. Give one situation where more friction is helpful.

4. Give one situation where less friction is helpful.

Round 2 - Sketch It

Draw the same block moving right on a smooth surface and a rough surface. Add friction arrows and compare their relative sizes.



Friction & Surface Type - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Rank likely friction from least to greatest: ice, smooth tile, carpet, sandpaper.

2. Why do athletic shoes often have textured soles?

3. Why does adding lubricant to machine parts reduce heating and wear?

Round 4 - Reason from Evidence

Friction can be both useful and harmful. Explain one example of each using force and motion language.

Friction & Surface Type - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Friction & Surface Type in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect friction to net force. If friction is the only horizontal force on a moving object, what happens to its motion?

E**Extend**

Design a fair test to compare friction on two surface types. Identify what you would keep the same and what you would measure.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Friction & Surface Type

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Opposite the direction of relative motion or attempted motion.
2. Rough surface.
3. Examples: shoe traction, tires gripping, brakes stopping, holding objects.
4. Examples: lubricated machine parts, skating, sliding an object more easily.

Sketch It - look for

Friction arrows point left. Rough-surface arrow should be larger than smooth-surface arrow for the same setup.

Apply It

1. Ice, smooth tile, carpet, sandpaper (general expectation).
2. To increase friction/traction with the ground.
3. It reduces friction between moving surfaces.

Reason from Evidence - look for

Helpful example should show friction providing needed grip/control. Harmful example should show friction opposing motion, causing wear, or converting useful mechanical energy to thermal energy.

ACE - Extend look-for

Keep object, mass, pulling method/angle, and distance constant; change surface type; measure force needed or stopping distance/time.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Distance-Time Graphs

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus On a distance-time graph, a flat line means stopped, a straight sloped line means constant speed, and a steeper slope means a faster speed.

Round 1 - Retrieve It

1. What does the horizontal axis usually represent?

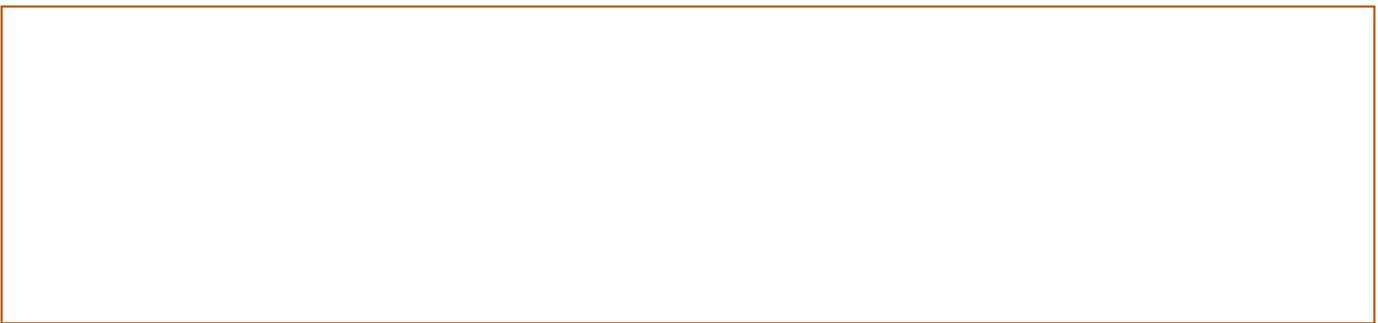
2. What does the vertical axis usually represent?

3. What does a flat segment mean?

4. How can you tell which of two straight segments represents faster motion?

Round 2 - Sketch It

Draw a distance-time graph for this trip: move steadily for 4 s, stop for 2 s, then move faster for 3 s. Label each segment.



Distance-Time Graphs - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A line goes from 0 m at 0 s to 20 m at 4 s. Average speed?

2. The graph stays at 20 m from 4 s to 6 s. What is the object doing?

3. From 6 s to 10 s, distance increases from 20 m to 50 m. Average speed on this segment?

Round 4 - Reason from Evidence

Compare the first and last moving segments in the practice data. Which is faster, and how do both the numbers and the graph show it?

Distance-Time Graphs - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Distance-Time Graphs in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect slope to speed. Why does a steeper distance-time line mean more distance is covered per unit time?

E**Extend**

Create a short story that matches a graph with a steep rise, a flat section, and then a gentle rise.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Distance-Time Graphs

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Time.
2. Distance from the starting point or distance traveled, depending on the graph.
3. Distance is not changing; the object is stopped.
4. The faster segment has the steeper slope.

Sketch It - look for

First rising straight line, then horizontal segment for 2 s, then a steeper rising line for 3 s.

Apply It

1. 5 m/s.
2. Stopped for 2 s.
3. $30 \text{ m} / 4 \text{ s} = 7.5 \text{ m/s}$.

Reason from Evidence - look for

Last segment is faster: 7.5 m/s vs 5 m/s, and its graph segment would be steeper.

ACE - Extend look-for

Accept stories that correctly map steep = fast, flat = stopped, gentle slope = slower constant motion.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Potential & Kinetic Energy

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Potential energy is stored energy due to position or condition, while kinetic energy is energy of motion; greater height generally means more gravitational potential energy.
------------------	---

Round 1 - Retrieve It

1. What is kinetic energy?

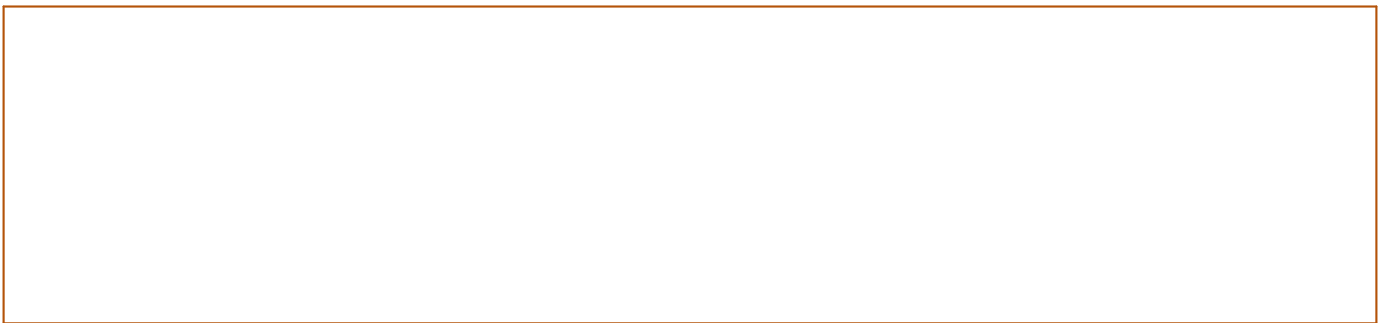
2. What is gravitational potential energy?

3. At which location on a hill is gravitational potential energy greatest, all else equal?

4. What happens to kinetic energy when an object speeds up, all else equal?

Round 2 - Sketch It

Draw a roller coaster at the top of a hill, halfway down, and near the bottom. Use PE and KE bars to show the energy trend.



Potential & Kinetic Energy - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A book sits on a high shelf. Which energy type is emphasized?

2. A rolling bowling ball is moving. Which energy type is emphasized?

3. Two identical balls move at different speeds. Which has more kinetic energy?

Round 4 - Reason from Evidence

Describe the energy transformation of a ball released from the top of a ramp. Include what happens to potential and kinetic energy.

Potential & Kinetic Energy - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Potential & Kinetic Energy in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect potential and kinetic energy to friction. Where can some mechanical energy go when friction is present?

E**Extend**

Predict how raising the starting point of a roller coaster affects its potential energy and possible speed later on the track.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Potential & Kinetic Energy

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Energy of motion.
2. Stored energy due to position/height in a gravitational field.
3. At the highest location.
4. It increases.

Sketch It - look for

Top: high PE, lower KE if slow. Downhill: PE decreases while KE increases. Bottom: lower PE and often higher KE, ignoring losses.

Apply It

1. Gravitational potential energy.
2. Kinetic energy.
3. The faster ball.

Reason from Evidence - look for

Gravitational potential energy decreases as height decreases; kinetic energy increases as speed increases. Some energy may transfer to thermal/sound due to friction.

ACE - Extend look-for

Higher starting point gives more gravitational potential energy and can allow greater kinetic energy/speed later, subject to energy losses.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Heat Transfer

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Thermal energy moves from warmer to cooler areas by conduction through direct contact, convection through fluid movement, and radiation by electromagnetic waves.
------------------	---

Round 1 - Retrieve It

1. What is conduction?

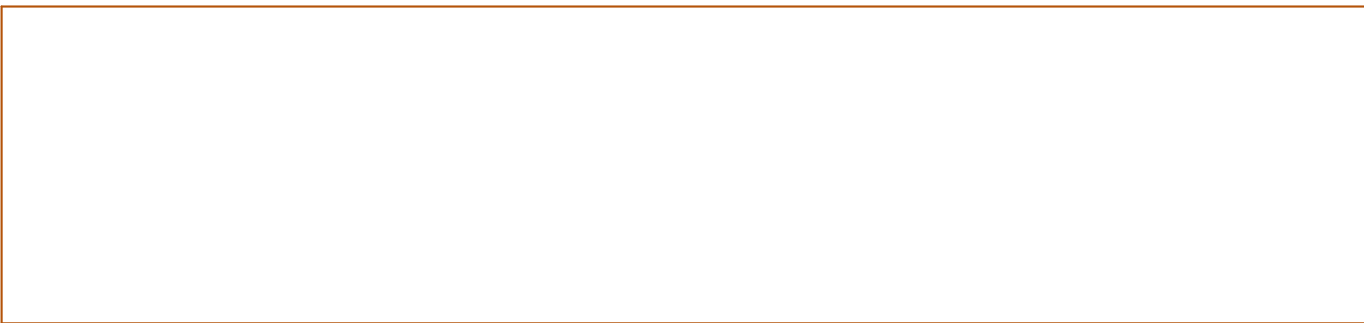
2. What is convection?

3. What is radiation?

4. In what overall direction does heat move spontaneously?

Round 2 - Sketch It

Draw a pot of water on a hot stove. Label one example each of conduction, convection, and radiation in or around the setup.



Heat Transfer - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A metal spoon gets hot while sitting in soup. Main heat-transfer method through the spoon?

2. Warm water rises and cool water sinks in a pot. Main method?

3. You feel warmth from the Sun across space. Main method?

Round 4 - Reason from Evidence

At a campfire, you may experience more than one type of heat transfer. Describe two methods and the evidence for each.

Heat Transfer - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Heat Transfer in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect heat transfer to particle motion. How does warmer matter differ from cooler matter at the particle level?

E**Extend**

Design one change to a cup that would slow heat transfer from a hot drink, and explain which transfer method your design reduces.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Heat Transfer

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Heat transfer through direct contact or particle collisions, especially in solids.
2. Heat transfer by bulk movement/circulation in liquids or gases.
3. Energy transfer by electromagnetic waves; no matter is required between source and receiver.
4. From warmer to cooler areas.

Sketch It - look for

Conduction through pot/handle or contact; convection currents in water/air; radiation from burner/flame/hot surfaces.

Apply It

1. Conduction.
2. Convection.
3. Radiation.

Reason from Evidence - look for

Radiation from fire to body; convection as hot air/smoke rises and moves; conduction if touching a heated object. Need mechanism-based explanation.

ACE - Extend look-for

Accept insulation, lid, reflective surface, reduced conduction/convection/radiation with clear reasoning.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Wave Properties

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Amplitude describes wave height from equilibrium, wavelength is the distance between matching points, and frequency is how many cycles pass per unit time; in the same medium, higher frequency corresponds to shorter wavelength.
------------------	--

Round 1 - Retrieve It

1. What does amplitude measure on a transverse wave?

2. What is wavelength?

3. What is frequency?

4. In the same medium at the same wave speed, how are frequency and wavelength related?

Round 2 - Sketch It

Draw two transverse waves in equal-width boxes: Wave A with larger amplitude and Wave B with shorter wavelength. Label amplitude and wavelength.

Wave Properties - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which wave has greater amplitude: one with 2 cm height from midline or 5 cm?

2. Two waves travel at the same speed. Wave X has higher frequency than Wave Y. Which has shorter wavelength?

3. For sound, what property most directly relates to pitch?

Round 4 - Reason from Evidence

Explain why changing frequency and wavelength in opposite directions is expected when wave speed stays constant.

Wave Properties - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Wave Properties in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect amplitude to energy. How might a larger-amplitude wave differ in effect from a smaller-amplitude wave?

E**Extend**

Create two wave descriptions that have the same frequency but different amplitudes. Predict one observable difference.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Wave Properties

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Maximum displacement from equilibrium; often shown as height from the midline to a crest or trough.
2. Distance between matching points on consecutive waves, such as crest to crest.
3. Number of wave cycles passing a point per unit time.
4. Higher frequency means shorter wavelength; lower frequency means longer wavelength.

Sketch It - look for

Wave A visibly taller from midline; Wave B has more cycles/shorter crest-to-crest spacing across same width; labels placed correctly.

Apply It

1. 5 cm.
2. Wave X.
3. Frequency.

Reason from Evidence - look for

Use relationship $\text{speed} = \text{frequency} \times \text{wavelength}$. If speed stays constant, increasing one factor requires decreasing the other.

ACE - Extend look-for

Same cycle rate/pitch (for sound) but different intensity/loudness or energy; accept other correct wave contexts.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Electromagnetic Spectrum

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	The electromagnetic spectrum runs from radio to gamma rays; as wavelength decreases, frequency and energy increase. Visible light is only a small portion.
------------------	--

Round 1 - Retrieve It

1. List the spectrum from longest wavelength to shortest.

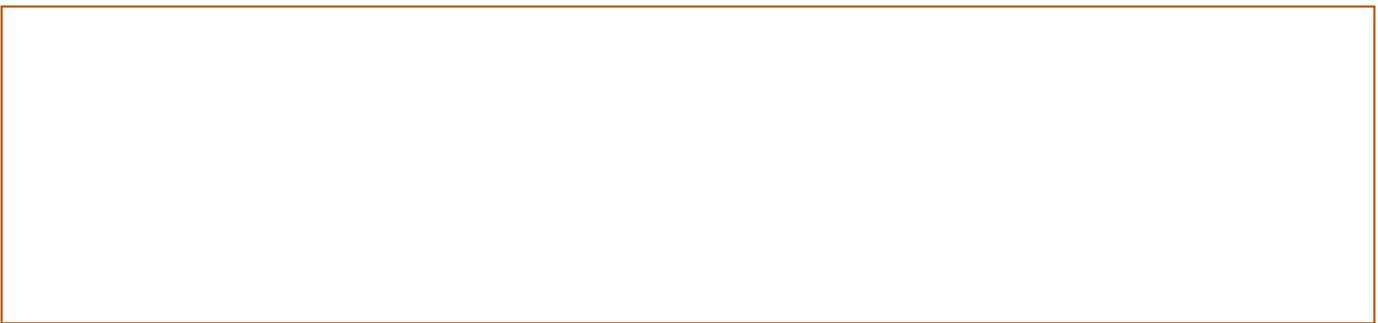
2. As frequency increases, what happens to wavelength?

3. As frequency increases, what generally happens to photon energy?

4. Is visible light most of the electromagnetic spectrum?

Round 2 - Sketch It

Draw a horizontal spectrum bar from radio to gamma. Add arrows showing wavelength decreases while frequency and energy increase.



Electromagnetic Spectrum - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which has longer wavelength: radio waves or X-rays?

2. Which has higher frequency: ultraviolet or infrared?

3. Match the use: thermal imaging/remote controls, bone imaging, communication broadcasts.

Round 4 - Reason from Evidence

Compare radio waves and gamma rays using wavelength, frequency, and energy.

Electromagnetic Spectrum - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Electromagnetic Spectrum in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect visible light to the full spectrum. Why can two electromagnetic waves behave differently even though both are electromagnetic radiation?

E**Extend**

A new detector measures radiation with a shorter wavelength than visible light but longer than X-rays. Which region is a likely possibility?

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Electromagnetic Spectrum

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Radio, microwave, infrared, visible, ultraviolet, X-ray, gamma.
2. It decreases.
3. It increases.
4. No. It is a small portion.

Sketch It - look for

Correct order with wavelength arrow opposite frequency/energy arrows.

Apply It

1. Radio waves.
2. Ultraviolet.
3. Infrared; X-rays; radio waves.

Reason from Evidence - look for

Radio: longest wavelength, lowest frequency/energy end. Gamma: shortest wavelength, highest frequency/energy end.

ACE - Extend look-for

Ultraviolet is the most direct answer in the standard order; depending on exact wavelength, boundary regions may vary.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Plate Boundaries

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Divergent boundaries move apart, convergent boundaries move together, and transform boundaries slide past each other; each pattern is linked to characteristic landforms and geologic activity.
------------------	---

Round 1 - Retrieve It

1. What happens at a divergent boundary?

2. What happens at a convergent boundary?

3. What happens at a transform boundary?

4. Which boundary types can produce earthquakes?

Round 2 - Sketch It

Draw three small boundary diagrams with arrows for divergent, convergent, and transform motion. Add one landform or event linked to each.

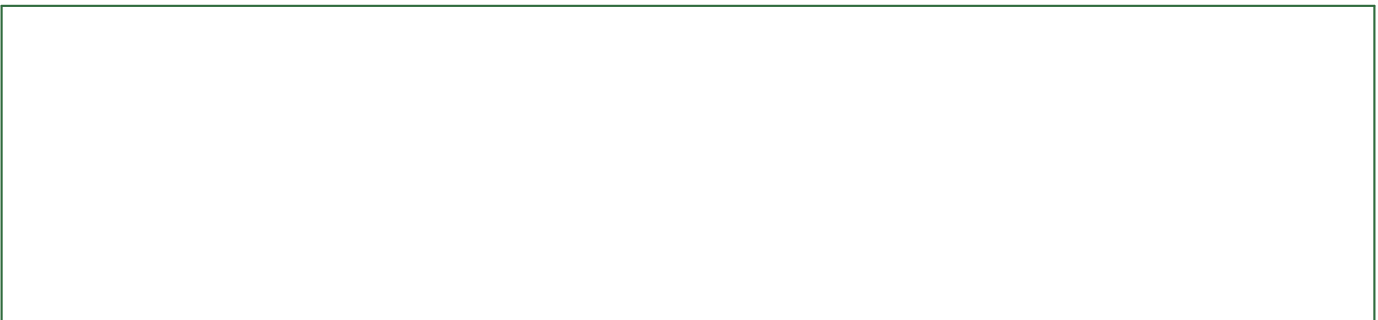


Plate Boundaries - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A mid-ocean ridge forms where seafloor spreads. Boundary type?

2. Two continental plates collide and build mountains. Boundary type?

3. Two plates slide past along a fault and produce earthquakes. Boundary type?

Round 4 - Reason from Evidence

Explain how the direction of plate motion helps you predict the kinds of geologic features that may form.

Plate Boundaries - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Plate Boundaries in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect plate boundaries to the rock cycle. How can plate movement contribute to melting, uplift, or metamorphism?

E**Extend**

A map shows a long ocean trench next to a volcanic arc. Infer the likely boundary type and explain your evidence.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Plate Boundaries

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Plates move apart.
2. Plates move toward each other.
3. Plates slide horizontally past each other.
4. All three can, though the settings and associated features differ.

Sketch It - look for

Divergent: arrows apart, ridges/rifts/new crust. Convergent: arrows together, mountains/trenches/volcanoes/subduction depending on plates. Transform: arrows sliding past, faulting/earthquakes.

Apply It

1. Divergent.
2. Convergent.
3. Transform.

Reason from Evidence - look for

Motion controls whether crust is created/separated, compressed/destroyed/thickened, or sheared sideways; those mechanisms produce different features.

ACE - Extend look-for

Convergent subduction boundary; trench and volcanic arc are common evidence of one plate sinking beneath another.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

The Rock Cycle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Igneous, sedimentary, and metamorphic rocks change over time through processes such as cooling, weathering/erosion, compaction/cementation, heat/pressure, melting, and uplift.
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Round 1 - Retrieve It

1. How does magma or lava become igneous rock?

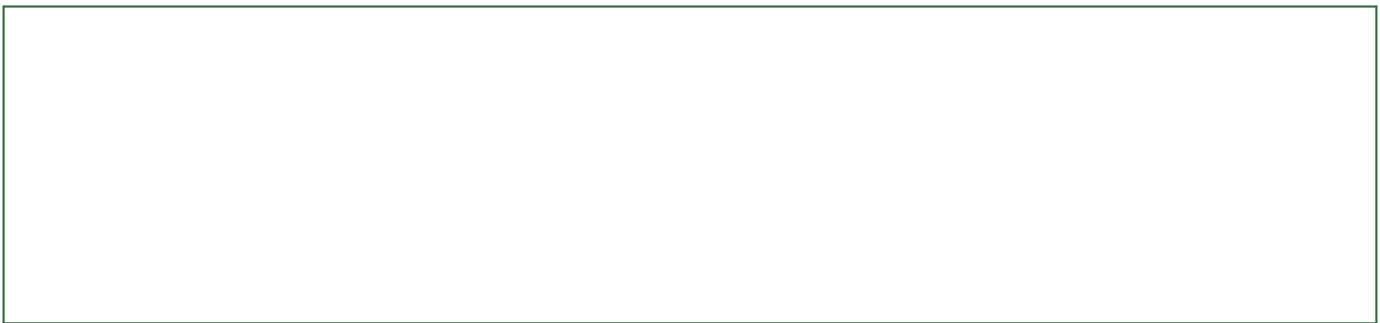
2. How do sediments become sedimentary rock?

3. How does existing rock become metamorphic rock?

4. What process can turn rock into magma?

Round 2 - Sketch It

Create a triangle with igneous, sedimentary, and metamorphic at the corners. Add arrows and process labels that show at least four ways rocks can change.



The Rock Cycle - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Granite melts underground. What material forms before a new rock type can form?

2. Sand is deposited, compacted, and cemented. What rock type forms?

3. Limestone is changed by heat and pressure without melting. What rock type results?

Round 4 - Reason from Evidence

Why is the rock cycle a cycle rather than a one-way sequence?

The Rock Cycle - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain The Rock Cycle in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect the rock cycle to plate tectonics. Which plate processes can expose, bury, melt, or deform rocks?

E**Extend**

Write one possible multi-step pathway that changes an igneous rock into a sedimentary rock and later into a metamorphic rock.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - The Rock Cycle

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. It cools and solidifies/crystallizes.
2. Compaction and cementation/lithification.
3. Heat and pressure change it without fully melting it.
4. Melting.

Sketch It - look for

Accurate processes such as cooling/crystallization, weathering/erosion/deposition, compaction/cementation, heat/pressure, melting, uplift.

Apply It

1. Magma.
2. Sedimentary.
3. Metamorphic.

Reason from Evidence - look for

Any rock type can follow multiple pathways and can change into several other types depending on processes and conditions; there is no single starting or ending point.

ACE - Extend look-for

Example: weathering/erosion -> sediments -> deposition/compaction/cementation -> sedimentary -> burial/heat/pressure -> metamorphic.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Weathering, Erosion & Deposition

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Weathering breaks rock/material into smaller pieces, erosion moves the material, and deposition drops or settles the material in a new location.
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Round 1 - Retrieve It

1. What does weathering do?

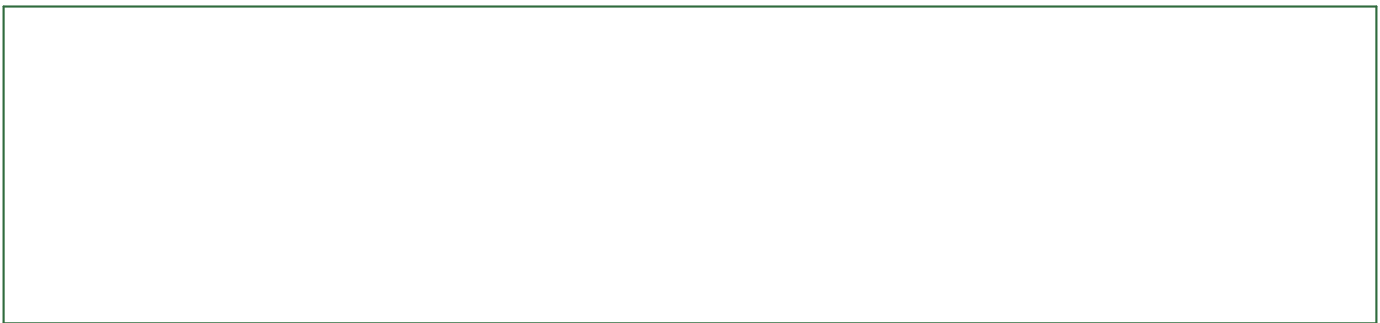
2. What does erosion do?

3. What does deposition do?

4. Name two agents that can cause erosion.

Round 2 - Sketch It

Draw a hillside-to-river scene. Label one place where weathering happens, one arrow showing erosion, and one area of deposition.



Weathering, Erosion & Deposition - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Tree roots crack a rock into smaller pieces. Which process?

2. A stream carries sand downstream. Which process?

3. Sediment settles where a river enters a lake. Which process?

Round 4 - Reason from Evidence

Describe a sequence in which one rock fragment experiences all three processes after a heavy rain.

Weathering, Erosion & Deposition - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Weathering, Erosion & Deposition in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect erosion and deposition to energy of moving water. What happens to sediment transport when water slows down?

E**Extend**

Predict how removing vegetation from a slope could change erosion after a storm, and explain why.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Weathering, Erosion & Deposition

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Breaks down rock or other material in place.
2. Moves weathered sediment/material.
3. Drops or settles transported sediment.
4. Examples: water, wind, ice/glaciers, gravity.

Sketch It - look for

Weathering at source rock/soil, movement by water/wind/gravity, deposition where energy decreases such as river bend/delta/bottom.

Apply It

1. Weathering.
2. Erosion.
3. Deposition.

Reason from Evidence - look for

Example: rain/roots break rock (weathering), runoff carries fragment downslope (erosion), water slows and fragment settles (deposition).

ACE - Extend look-for

Erosion likely increases because roots/plant cover no longer hold soil or reduce runoff impact; more sediment may be transported and deposited elsewhere.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Weather Systems & the Water Cycle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Solar energy drives evaporation; water vapor condenses into clouds, falls as precipitation, and returns through runoff, infiltration, and collection before cycling again.
------------------	--

Round 1 - Retrieve It

1. What process changes liquid water to water vapor?

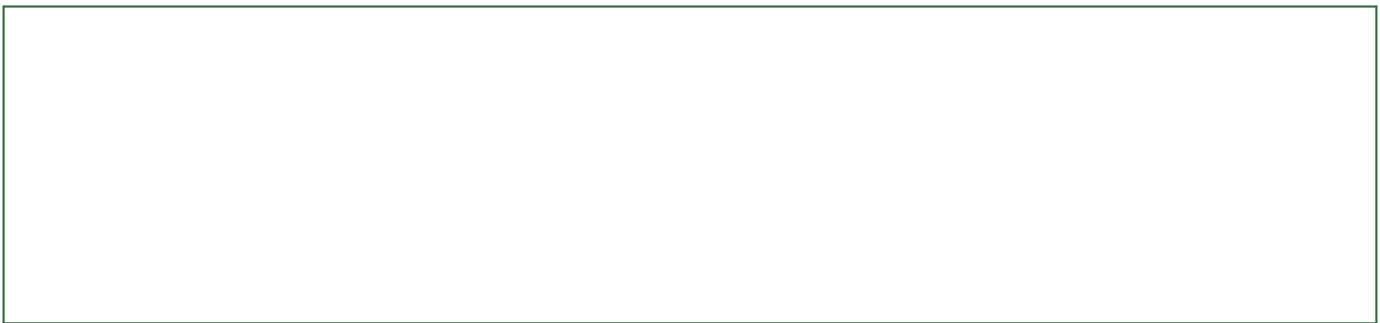
2. What process changes water vapor to liquid droplets?

3. What is precipitation?

4. What major energy source drives the water cycle?

Round 2 - Sketch It

Draw a water-cycle diagram with evaporation, condensation, precipitation, runoff, and collection. Add the Sun as the energy source.



Weather Systems & the Water Cycle - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A puddle becomes smaller on a warm day. Which process dominates?

2. Water droplets form on the outside of a cold cup. Which process?

3. Rainwater flows over land into a creek. Which process is being described?

Round 4 - Reason from Evidence

Trace one water molecule from the ocean into a cloud, onto land, and back toward the ocean. Name the processes along the way.

Weather Systems & the Water Cycle - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Weather Systems & the Water Cycle in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect the water cycle to weather. Why are condensation and cloud formation important to precipitation?

E**Extend**

Predict how several very hot, dry days could affect evaporation from a shallow pond and the pond's water level.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Weather Systems & the Water Cycle

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Evaporation.
2. Condensation.
3. Water that falls from clouds, such as rain, snow, sleet, or hail.
4. The Sun.

Sketch It - look for

Correct process order/relationships, with water moving between surface and atmosphere; Sun linked to evaporation.

Apply It

1. Evaporation.
2. Condensation of water vapor from air.
3. Runoff.

Reason from Evidence - look for

Example: evaporation -> condensation -> precipitation -> runoff/groundwater/collection -> ocean.
Transpiration/infiltration can be included if accurately used.

ACE - Extend look-for

Evaporation rate likely increases, lowering pond level if inputs do not replace the lost water.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Phases of the Moon

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus The eight moon phases repeat in order as the Moon orbits Earth; waxing means the illuminated portion we see is growing, while waning means it is shrinking.

Round 1 - Retrieve It

1. What does waxing mean?

2. What does waning mean?

3. What phase comes after new moon?

4. What phase comes after full moon?

Round 2 - Sketch It

Make an eight-position phase sequence and label: new, waxing crescent, first quarter, waxing gibbous, full, waning gibbous, third quarter, waning crescent.

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Phases of the Moon - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which phase comes between waxing crescent and waxing gibbous?

2. Which phase comes between waning gibbous and waning crescent?

3. Does the Moon produce its own visible light?

Round 4 - Reason from Evidence

Explain why ordinary moon phases are not caused by Earth's shadow falling on the Moon.

Phases of the Moon - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Phases of the Moon in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect phases to the Moon's orbit. Why does the sequence repeat?

E**Extend**

If the Moon is at first quarter tonight, predict the general trend in visible illumination over the next several days.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Phases of the Moon

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The visible illuminated portion is increasing.
2. The visible illuminated portion is decreasing.
3. Waxing crescent.
4. Waning gibbous.

Sketch It - look for

Correct cyclic order. Exact shading orientation depends on viewing convention/hemisphere; phase names and waxing/waning sequence matter most.

Apply It

1. First quarter.
2. Third quarter.
3. No. We see sunlight reflected from the Moon.

Reason from Evidence - look for

Phases result from changing viewing geometry of the Moon's sunlit half as it orbits Earth. Earth's shadow causes a lunar eclipse, not the regular monthly phases.

ACE - Extend look-for

It will continue waxing toward waxing gibbous and then full moon.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Star Life Cycle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Stars form from nebulae, spend most of their lives in the main-sequence stage, and follow different end paths depending largely on mass.
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Round 1 - Retrieve It

1. What is a nebula?

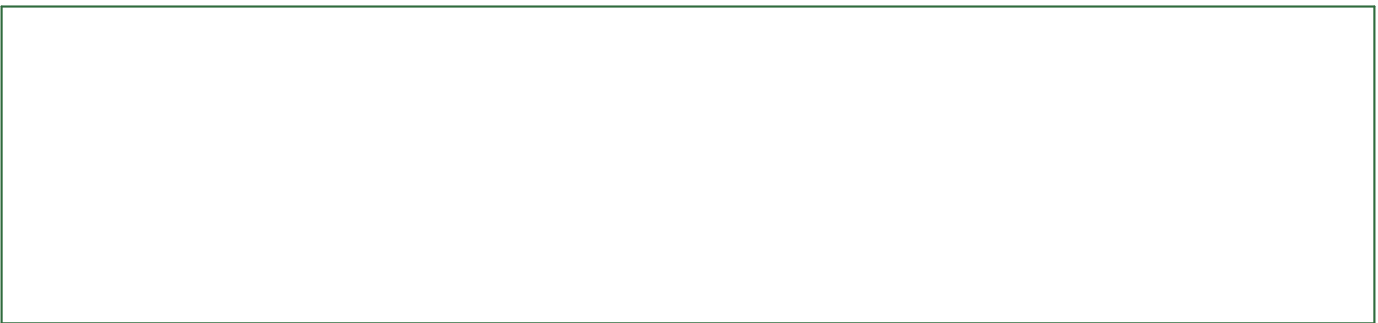
2. What stage follows a protostar once stable fusion is established?

3. What is a common end product for a Sun-like star?

4. What major event can end the life of a massive star?

Round 2 - Sketch It

Draw two branching pathways from nebula to main sequence: one for a Sun-like star and one for a massive star.



Star Life Cycle - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Which factor most strongly determines which life-cycle path a star follows?

2. What comes before a white dwarf in a Sun-like star path?

3. What can remain after a supernova?

Round 4 - Reason from Evidence

Explain why the star life cycle is better shown as a branching diagram than as one single line.

Star Life Cycle - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Star Life Cycle in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect star mass to gravity. Why might a more massive star follow a more extreme end path?

E**Extend**

Predict the broad end path of the Sun based on its being a lower-mass/main-sequence star rather than a very massive star.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Star Life Cycle

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A cloud of gas and dust where stars can form.
2. Main-sequence star.
3. White dwarf, after red giant and planetary-nebula stages.
4. A supernova, which may leave a neutron star or black hole.

Sketch It - look for

Both begin nebula -> protostar -> main sequence. Sun-like: red giant -> planetary nebula -> white dwarf. Massive: red supergiant -> supernova -> neutron star or black hole.

Apply It

1. Mass.
2. Planetary nebula after the red giant phase.
3. A neutron star or black hole, depending on mass/remnant conditions.

Reason from Evidence - look for

Stars do not all follow the same ending; mass leads to different stages and remnants, so the sequence branches.

ACE - Extend look-for

Red giant -> planetary nebula -> white dwarf; not a core-collapse supernova/black hole path.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Cell Structures

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Key cell structures include cell membrane, cell wall, nucleus, ribosomes, cytoplasm, mitochondria, chloroplasts, and vacuoles, each with a distinct function.
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Round 1 - Retrieve It

1. What does the cell membrane do?

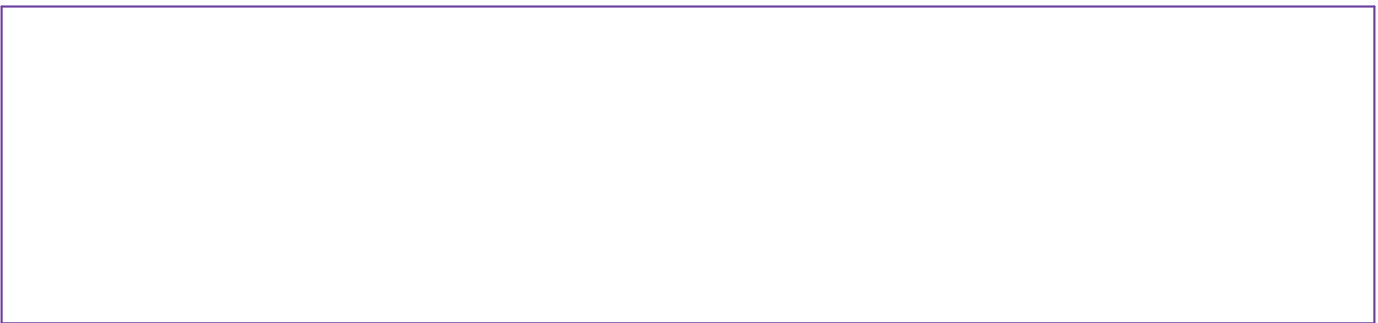
2. What does the nucleus do?

3. What do ribosomes do?

4. Which two structures are strongly associated with plant cells and not animal cells?

Round 2 - Sketch It

Draw a simple plant cell and animal cell. Label membrane, nucleus, cytoplasm, mitochondria, ribosomes, and vacuole. Add wall and chloroplasts to the plant cell.



Cell Structures - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A cell needs to make many proteins. Which structure is directly responsible?

2. A leaf cell captures light energy. Which structure is central to that job?

3. A muscle cell needs large amounts of usable energy. Which organelle would likely be abundant?

Round 4 - Reason from Evidence

Compare plant and animal cells using at least two similarities and two differences.

Cell Structures - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Cell Structures in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect cell structure to function. Why would different cell types have different numbers of mitochondria or chloroplasts?

E**Extend**

A mystery cell has a cell wall, chloroplasts, and a large vacuole. Identify the cell type and defend your claim.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Cell Structures

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Controls what enters and leaves the cell; forms a boundary.
2. Contains genetic material and helps direct cell activities.
3. Build proteins.
4. Cell wall and chloroplasts; plants also often have a large central vacuole.

Sketch It - look for

Correct placement does not need microscopic precision. Plant should include cell wall and chloroplasts; both cells include membrane, nucleus, cytoplasm, mitochondria, ribosomes, vacuoles.

Apply It

1. Ribosomes.
2. Chloroplasts.
3. Mitochondria.

Reason from Evidence - look for

Similarities can include membrane, nucleus, cytoplasm, mitochondria, ribosomes, vacuoles. Differences: plant cell wall, chloroplasts, typically larger central vacuole and more rigid shape.

ACE - Extend look-for

Plant cell; those structures are characteristic of plant cells.

Suggested use

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Photosynthesis

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus Photosynthesis uses sunlight, carbon dioxide, and water to produce glucose and oxygen:
 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.

Round 1 - Retrieve It

1. What are the three main inputs for photosynthesis?

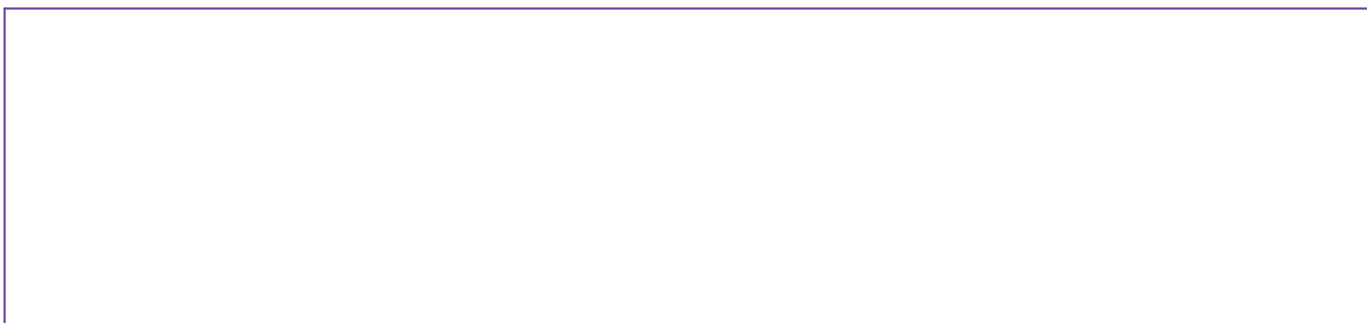
2. What are the main products?

3. In which organelle does photosynthesis occur in plant cells?

4. What form of stored chemical energy is produced?

Round 2 - Sketch It

Draw a leaf with arrows showing sunlight, CO_2 , and H_2O going in and glucose and O_2 coming out. Label the chloroplast as the cellular site.



Photosynthesis - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Count atoms on each side of $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Is the equation balanced?

2. A plant is kept in darkness but still has water and CO_2 . Which required input is missing?

3. Why is glucose important to the plant?

Round 4 - Reason from Evidence

Explain how photosynthesis connects energy from the Sun to consumers in a food web.

Photosynthesis - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Photosynthesis in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect photosynthesis to gas exchange. Which gas is taken in and which is released during photosynthesis?

E**Extend**

Predict how reducing available light for several days could affect a plant's glucose production, assuming other factors stay similar.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Photosynthesis

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Light energy, carbon dioxide, and water.
2. Glucose and oxygen.
3. Chloroplast.
4. Glucose/sugar.

Sketch It - look for

Inputs and outputs oriented correctly; chloroplast identified; glucose shown as product/stored chemical energy.

Apply It

1. Yes: C = 6, H = 12, O = 18 on both sides.
2. Light energy.
3. It stores chemical energy and provides carbon/building material for growth and metabolism.

Reason from Evidence - look for

Producers capture light energy and store it in glucose/biomass; consumers obtain chemical energy by eating producers or organisms that ate producers.

ACE - Extend look-for

Glucose production should decrease because light energy is required to drive photosynthesis.

Suggested use

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Food Chains & Food Webs

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Food chains show one pathway of energy flow, while food webs show many connected feeding pathways; arrows point in the direction energy moves.
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Round 1 - Retrieve It

1. What does a food chain show?

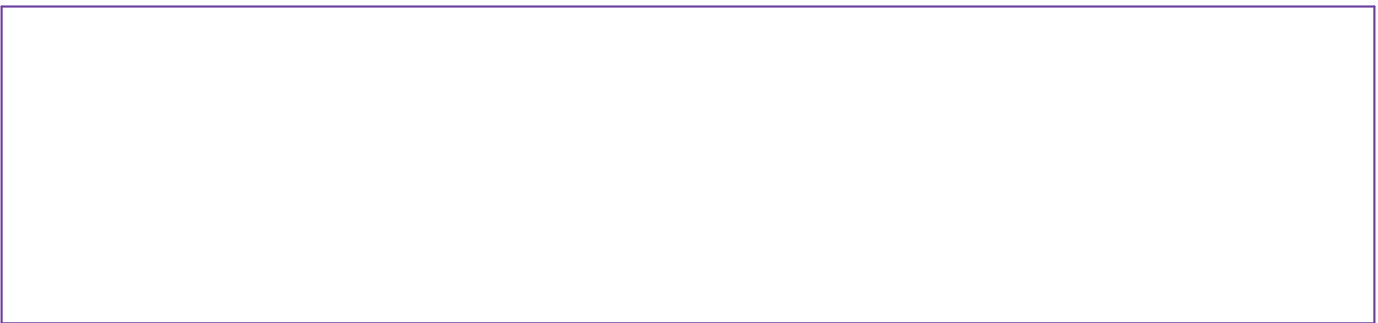
2. What does a food web show?

3. Which direction should an energy arrow point?

4. What is a producer?

Round 2 - Sketch It

Build a food web using grass, grasshopper, rabbit, frog, snake, and hawk. Draw arrows to show energy flow. Include at least five arrows.



Food Chains & Food Webs - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. In grass → rabbit → fox, where does the fox's energy ultimately begin?

2. Why is a food web usually more realistic than one food chain?

3. If a producer population drops sharply, which groups may be affected?

Round 4 - Reason from Evidence

A predator disappears from a food web. Explain why predicting the outcome requires considering more than one chain.

Food Chains & Food Webs - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Food Chains & Food Webs in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect food webs to photosynthesis. Why are producers the entry point for most ecosystem energy?

E**Extend**

Choose one organism in your drawn web and predict two possible effects if its population is cut in half.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Food Chains & Food Webs

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Retrieve It

1. One pathway of energy transfer through organisms.
2. Many interconnected food chains/feeding relationships.
3. From the organism being eaten/energy source to the organism receiving the energy.
4. An organism that makes its own food, typically using photosynthesis.

Sketch It - look for

Arrows point from food/resource to consumer. Many valid webs exist if relationships are ecologically plausible and arrow direction is correct.

Apply It

1. With the Sun captured by the grass through photosynthesis.
2. Most organisms have multiple food sources and predators, so energy pathways are interconnected.
3. Primary consumers first, then higher-level consumers; effects can spread through the web.

Reason from Evidence - look for

Species have multiple links. Removing one predator can increase some prey, alter competition, affect producers, and shift other predators; web interactions matter.

ACE - Extend look-for

Answers should trace at least two direct or indirect energy/feeding relationships with plausible directions.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.

Ecological Succession

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Ecological succession is the gradual change in community composition over time; primary succession can begin on bare rock, while later stages often support greater soil development, plant structure, and biodiversity.
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Round 1 - Retrieve It

1. What is ecological succession?

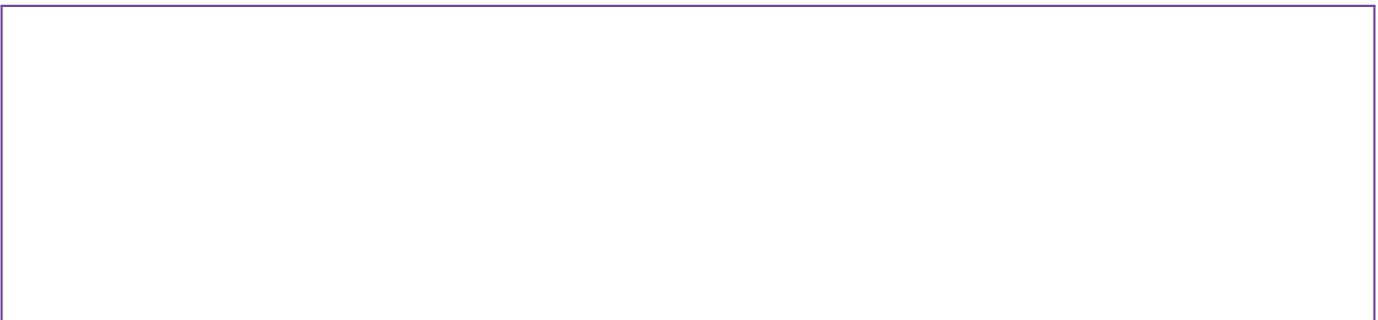
2. What kind of surface can primary succession begin on?

3. What are pioneer species?

4. What general biodiversity trend often occurs as succession proceeds?

Round 2 - Sketch It

Draw four stages from bare rock to a mature community. Include pioneer species, grasses/small plants, shrubs, and larger plants/trees where appropriate.



Ecological Succession - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. New lava cools into bare rock with no soil. Primary or secondary succession?

2. A wildfire removes vegetation but leaves soil. Primary or secondary succession?

3. Why are lichens often used as an example of pioneer species on bare rock?

Round 4 - Reason from Evidence

Compare what makes primary succession usually slower than secondary succession.

Ecological Succession - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Ecological Succession in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect succession to disturbance. How does the amount of soil and surviving life after a disturbance affect recovery?

E**Extend**

A construction site is abandoned after topsoil remains but most plants are removed. Predict the likely succession type and early changes.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Ecological Succession

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A gradual change in the species/community structure of an ecosystem over time.
2. Bare rock or another surface with little/no soil.
3. Early colonizers that can live in harsh conditions and help change the environment for later species.
4. Biodiversity and structural complexity often increase, though real ecosystems can vary.

Sketch It - look for

A reasonable progression showing soil/community development and increasing complexity. Exact species depend on ecosystem.

Apply It

1. Primary succession.
2. Secondary succession.
3. They can colonize harsh surfaces and help weather rock/build organic material, supporting soil development.

Reason from Evidence - look for

Primary starts without developed soil/seed bank and must build conditions from bare substrate. Secondary begins with soil and often surviving organisms/seeds, speeding recovery.

ACE - Extend look-for

Secondary succession; soil remains, so grasses/weeds and other early colonizers can establish relatively quickly, followed by later community changes.

Suggested use

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